CH 221 – Organic Chemistry I
Fall 2022
Section 004: TH 3:00 – 4:15 Dabney 124

Dr. Cassie Lilly
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Office/Student Hours: W, F 10 a.m – 11 a.m.

Course Materials

Textbook	***Organic Chemistry*** by F. A. Carey and R.M. Giuliano, 11th Edition. There are many purchase options for the textbook (new, used, ebook, rental, etc.)The expectation is that have already read the chapter assigned **before** coming to class. If you already have or can get another textbook, most organic chemistry textbooks will work. The course schedule will be based on sections from Carey, F. A., *Organic Chemistry*, 11th ed. Other acceptable texts include: Carey, F. A., *Organic Chemistry*, 7th - 10th ed.; Bruice, P. Y., *Organic Chemistry*, 7th - 8th ed.; Wade, L.G., *Organic Chemistry*, 7th - 9th ed.; Brown, W.H.; Foote, C.S.; Iverson, B.L.; Anslyn, E. *Organic Chemistry*, 5th - 8th ed.
Solutions Manual	Purchase the solutions manual which accompanies your textbook.
Molecular Modeling Kit	An organic chemistry model kit is highly recommended (for example: Chem-Tutor by Sigma-Aldrich, MolyMod, or MegaMolecules)
Moodle	Course materials such as notes, worksheets, quiz/test keys, extra resources will be posted on Moodle. Access the LMS through http://wolfware.ncsu.edu/
Achieve (Online HW)	You must register for the online homework platform Achieve Essentials for Organic Chemistry (1-term, ISBN:9781319392017) by Macmillian Learning to complete homework assignments. This is accessed by a link through the Moodle site.

Laboratory

CH 222 is the laboratory course which should be taken concurrently with CH 221 and is administered separately. Refer ***all*** inquiries about CH222 Laboratory to Dr. Cassie Lilly (cassie_lilly@ncsu.edu) 254 Fox Hall.

Grading

Percentage of Overall Grade

Homework	10 %
Quizzes (3 in total)	15 %
Exams (3)	55 %
Final Exam	20 %

The standard 10 point scale used by the University will be used to assign letter grades. The university grading regulation is located at http://policies.ncsu.edu/regulation/reg-02-50-03.

Homework: 10% of your course grade will come from homework delivered through Achieve. HW assignments will be posted before the first day of class therefore ***no extensions*** will be granted for reasons that would normally be considered as unexcused absences. The lowest HW grade will be dropped to account to technical difficulties and missed homework.

Quizzes: Quizzes will be administered during class on the days scheduled on Moodle.

Extra Credit/Curves: If extra credit is given for any assignment, the maximum score possible for that assignment will be a 100. Any "excess" extra credit points will not be applied to other assignments. ***There will be no curve applied to ANY assignment.***

Regrading Exams: If you request a regrade, the ENTIRE exam will be regraded at the discretion of the instructor, your score can increase or decrease. You have one week from the date of exams grades being recorded in Moodle to get your exam regraded.

Attendance and Missed Quizzes/Examinations: Be aware that missing lecture will put you at a serious disadvantage with respect to learning course material. If you will be absent on a quiz or exam day due to a preplanned event, you must notify me **at least one week** prior to that date and the quiz/exam will be administered ***EARLY***. If a quiz is missed for a University excused absence (http://policies.ncsu.edu/regulation/reg-02-20-03), that quiz will be made up during office hours within 7 days of the absence. If an exam is missed for a University excused absence (http://policies.ncsu.edu/regulation/reg-02-20-03), the student must notify the instructor as soon as possible to determine what course of action will be taken to makeup the absence. It may be that a makeup exam is not given. More than one excused exam absence or an excused absence for the final exam will result in an IN for the class. Information regarding university considered excused absences is given at: http://policies.ncsu.edu/regulation/reg-02-20-03. If an exam is cancelled due to adverse weather, the exam will be administered during the next class meeting.

Course Overview

Prerequisites: CH 101 with a grade of C- or better and CH 102**; Corequisite:** CH 222

Course Schedule: The course schedule is in the table below. Please note that this is a tentative schedule of the topics that could be adjusted as needed throughout the semester. The final exam date can be found here: https://studentservices.ncsu.edu/calendars/exam/#fall

Month	Day	Subject	Readings in Carey	Extra Problems (Carey 11th ed)
Aug	23	Introduction, Review: Lewis Structures, Electronegativity, Formal Charges; Resonance; Molecular Structure	1.1 – 1.9	
Aug	25	Using Curved Arrows in Chemical Reaction Mechanisms, Acids and Bases, How Structure Effects Acidity, pK_a and Equilibria, Lewis Acids and Bases	1.10 – 1.15	Chp 1 - 43,46,48-51,53,55,57-59,62,63,65-68
Aug	30	Valence Bond Theory vs Molecular Orbital Theory, Covalent Bonding and Hybridization, Bonding in Alkanes, Alkenes and Alkynes, Molecular Orbitals and Bonding in Methane	2.1 – 2.11	
Sept	1	Isomers, Higher Alkanes, Naming, Cycloalkanes, Occurrence and Production, Physical Properties, Combustion, Oxidation/Reduction	2.12 – 2.24	Chp 2 – 25-28-30,32, 35, 38
Sept	6	Conformations of Alkanes, Shapes of Cycloalkanes, Conformations of Cyclohexanes	3.1 – 3.7	
Sept	8	Axial and Equatorial Bonds, Cis/Trans Isomers, Conformational Analysis of Substituted Cyclohexanes and Other Ring Systems, Polycyclic Ring Systems, Heterocyclic Compounds	3.8 – 3.15	Chp 3 - 23-25,25,28,30,31,34-37,42,43
Sept	13	Chirality, Optical Activity, Relative and Absolute Configurations, R/S Notation, Fischer Projections	4.1 – 4.7	
Sept	15	Properties of Enantiomers, Chirality Axis, Molecules with Two Chirality Centers, Diastereomers, Meso Isomers	4.8 – 4.11	Chp 4: 29-31,35-40
Sept	20	**Exam #1**	**Chapters 1-4**	
Sept	22	Functional Groups, Naming Alkyl Halides and Alcohols, Bonding, Physical Properties, Synthesis of Alkyl Halides from Alochols	5.1 – 5.7	
Sept	27	S_N1 Mechanism , Potential Energy Diagrams, Carbocations, Reaction Rate Effects, Stereochemistry of Reactions Carbocation Rearrangements, SN2 Reaction, Other Reactions to Make Alkyl Halides From Alcohols, Sulfonates	5.8 – 5.15	Chp 5 - 19,20,32-36,41,43,44
Sept	29	Functional Group Transformation by Nucleophilic Substitution, Relative Reactivity, S_N2 Mechanism, Steric Effects, Nucleophiles and Nucleophilicity	6.1 – 6.5	

Oct	4	S_N1 Reaction, Solvent Effects, Nucleophilic Substitution of Alkyl Sulfonates, Retrosynthetic Analysis, Substitution vs Elimination	6.6 – 6.12	Chp 6 - 19-23,26,27,32,33,35
Oct	6	Naming and Isomers of Alkenes, Physical Properties, Stabilities of Alkenes and Cycloalkenes, Making Alkenes By Elimination, Alcohol Dehydration Mechanism	7.1 – 7.12	
Oct	11	**No Class- Fall Break**		
Oct	13	Rearrangements, Dehydrohalogenation, E2 Mechanism, Isotope Effects, E1 Mechanism, Substitution vs Elimination, Elimination of Sulfonates	7.13 – 7.20	Chp 7 - 31,39,41,42,43,45,49,50-55
Oct	18	**Exam #2**	**Chapters 5-7**	
Oct	20	Hydrogenation of Alkenes, Electrophilic Addition of Hydrogen Halides, Acid-Catalyzed Hydration	8.1 – 8.6	
Oct	25	Thermodynamics of Addition-Elimination Equilibria, Hydroboration-Oxidation, Halogen Addition, Epoxidation, Ozonolysis, Retrosynthetic Analysis	8.7 – 8.14	Chp 8 - 25,28-31,34,46-50,64-66
Oct	27	Alkyne Naming, Bonding, Properties, Acidity, Making Alkynes	9.1 – 9.7	
Nov	1	Alkyne Reactions: Hydrogenation, Addition of Hydrogen Halides, Hydration, Addition of Halogens, Ozonolysis, Alkynes in Synthesis	9.8 – 9.14	Chp 9 - 17,20,24,25,27,30-32,34,36
Nov	3	Structure, Bonding and Stability of Alkyl Radicals, Halogenation of Alkanes, Mechanism of Radical Chain Reactions	10.1 – 10.4	
Nov	8	Free-radical Addition of HBr to Alkenes, Metal-Ammonia Reduction of Alkynes, Free-Radical Polymerization of Alkenes	10.5 – 10.8	Chp 10 – 19, 20,21,23,25,29,30,32,33
Nov	10	Ethers	17.1, 17.4-17.6	
Nov	**15**	**Exam #3**	**Chapters 8-10**	
Nov	17	Epoxides	17.9-17.13	Chp 17 - 26,27,34,35,41,42
Nov	22	IR Spectroscopy	14.20-14.22	
Nov	24	**No Class-Thanksgiving**		
Nov	29	^{1}H-NMR Interpretation	14.3-14.11	
Dec	1	^{1}H-NMR and ^{13}C-NMR	14.12-14.16	Chp 14 - 31,32,36,39,41,46,48,50,53
Dec	**13**	**Final Exam at 3:30 p.m.**	**Comprehensive**	

Course Description and Student Learning Outcomes:

Description: First half of two-semester sequence in the fundamentals of modern organic chemistry. Structure and bonding, stereochemistry, reactivity and synthesis of carbon compounds. Detailed coverage of aliphatic hydrocarbons, alcohols, ethers, and alkyl halides. Introduction to spectral techniques of IR, UV-vis, and NMR.

Learning Outcomes:

1. Draw structures and name various organic compounds including:
   A. alkanes, alkenes, alkynes
   B. alcohols
   C. alkyl halides
   D. ethers & sulfides
2. Draw structures and name organic compounds that have configurations including
   A. E/Z
   B. R/S
3. Describe the physical properties of the compounds listed in #1 above including:
   A. bonding
   B. boiling point trends
   C. heats of combustion/hydrogenation trends
   D. solubility trends
4. Compare the relative stabilities of various molecules including:
   A. cyclohexane derivatives
   B. hydrocarbon isomers
5. Describe the proposed mechanism for various reactions by drawing the electron flow arrows, transition states and the resulting intermediates and products including:
   A. S_N1
   B. S_N2
   C. E1
   D. E2
   E. Free radical reactions
   F. Electrophilic addition
   G. Metal catalyzed hydrogenation
6. Devise synthetic schemes for preparing various organic compounds including:
   A. alkanes
   B. alkenes
   C. alkynes
   D. alcohols
   E. alkyl halides
   F. ethers & sulfides (including epoxide chemistry)
7. Predict the products for the reactions of various functional groups including:
   A. alkanes
   B. alkenes
   C. alkynes
   D. alcohols
   E. alkyl halides
   F. ethers & sulfides (including epoxide chemistry)
   G. allylic systems
   H. conjugated dienes

8. Predict the products for reactions that are:
    A. stereospecific
    B. stereoselective
    C. regiospecific
    D. regioselective
9. Describe the property of chirality and how it applies to:
    A. reactions
    B. drugs
10. Use spectroscopy to elucidate structures by interpreting the spectra generated from:
    A. ^{1}H and ^{13}C NMR spectroscopy
    B. IR spectroscopy
    C. UV-Vis spectroscopy

Course Status

Audits: To obtain a grade of AU in the course, the student must take ALL exams. Information about and requirements for auditing a course can be found at http://policies.ncsu.edu/regulation/reg-02-20-04.

Credit Only: In order to receive a grade of S, students are required to take all exams and quizzes, complete all assignments, and earn a grade of C- or better. Conversion from letter grading to credit only (S/U) grading is subject to university deadlines. Refer to the Registration and Records calendar for deadlines related to grading. For more details refer to http://policies.ncsu.edu/regulation/reg-02-20-15.

Changing your Status (or dropping): ***Friday, August 26th is the last day to drop a course without a W grade.*** ***Wednesday, October 19th is the last day to change to credit only and the last day to drop with a W grade.*** After this date, modifications to your status in this class are determined by the COS Late Schedule Revision Policy. A full description of this policy can be found at the following website; http://sciences.ncsu.edu/students/academic-changes/late-changes/. Dropping specific courses (or changing to CR or AU) after this deadline is considered only for unforeseen and unavoidable extenuating personal situations (***not poor performance***).

https://studentservices.ncsu.edu/calendars/academic/#fall

University Policies

Incomplete Grades: If an extended deadline is not authorized by the instructor or department, an unfinished incomplete grade will automatically change to an F after either (a) the end of the next regular semester in which the student is enrolled (not including summer sessions), or (b) the end of 12 months if the student is not enrolled, whichever is shorter. Incompletes that change to F will count as an attempted course on transcripts. The burden of fulfilling an incomplete grade is the responsibility of the student. The university policy on incomplete grades is located at http://policies.ncsu.edu/regulation/reg-02-50-03.

Responsible Conduct: Students are required to comply with the university policy on academic integrity found in the Code of Student Conduct found at http://policies.ncsu.edu/policy/pol-11-35-01. See http://policies.ncsu.edu/policy/pol-11-35-01 for a detailed explanation of academic honesty. Your signature on any test or assignment indicates "I have neither given nor received unauthorized aid on this test or assignment."

Accommodations for Disabilities: Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with the Disability Resource Office at Holmes Hall, Suite 304, 2751 Cates Avenue, Campus Box 7509, 919-515-7653. For more information on NC State's policy on working with students with disabilities, please see the Academic Accommodations for Students with Disabilities Regulation (REG02.20.01).

Supporting Fellow Students in Distress: As members of the NC State Wolfpack community, we each share a personal responsibility to express concern for one another and to ensure that the classroom and the campus as a whole remains a safe environment for learning. Occasionally, you may come across a fellow classmate whose personal behavior concerns or worries you. If this is the case, you are encouraged to report this behavior to the NC State Cares website. Although you can report anonymously, it is preferred that you share your contact information so they can follow-up with you personally. Students are responsible for reviewing the NC State University PRR's which pertains to their course rights and responsibilities:

- Equal Opportunity and Non-Discrimination Policy Statement
- Code of Student Conduct
- Grades and Grade Point Average

Health and Well-Being Resources

Sometimes during a student's undergraduate career, academic and personal stress is a natural result. All students are encouraged to take care of themselves and their peers. If you need additional support, there are many resources on campus to help you:

- Counseling Center (https://counseling.dasa.ncsu.edu/)
- Health Center (https://healthypack.dasa.ncsu.edu/)
- If you or someone you know are experiencing food, housing or financial insecurity, please see the Pack Essentials Program (https://dasa.ncsu.edu/pack-essentials/).

COVID-19 Syllabus Content

Due to the COVID-19 pandemic, public health measures continue to be implemented across campus. Students should stay current with these practices and expectations through the Protect the Pack website (https://www.ncsu.edu/coronavirus/). The sections below provide expectations and conduct related to COVID-19 issues.

Health and Participation in Class

We are most concerned about your health and the health of your classmates and instructors/TAs.

- If you test positive for COVID-19, or are told by a healthcare provider that you are presumed positive for the virus, you should not attend any hybrid or face-to-face (F2F) classes and work with your instructor on any adjustments necessary; also follow other university guidelines, including self reporting (Coronavirus Self Reporting): Self-reporting is not only to help provide support to you, but also to assist in contact tracing for containing the spread of the virus.

- If you feel unwell, even if you have not been knowingly exposed to COVID-19, please do not come to a F2F class or activity.
- If you are in quarantine, have been notified that you may have been exposed to COVID-19, or have a personal or family situation related to COVID-19 that prevents you from attending this course in person (or synchronously), please connect with your instructor to make alternative plans, as necessary.
- If you need to make a request for an academic consideration related to COVID-19, such as a discussion about possible options for remote learning, please talk with your instructor.

Health and Well-Being Resources

These are difficult times, and academic and personal stress are natural results. Everyone is encouraged to take care of themselves and their peers. If you need additional support, there are many resources on campus to help you:

- Counseling Center (NCSU Counseling Center)
- Student Health Services (Health Services | Student)
- If the personal behavior of a classmate concerns or worries you, either for the classmate's well-being or yours, we encourage you to report this behavior to the NC State CARES team: (Share a Concern).
- If you or someone you know are experiencing food, housing or financial insecurity, please see the Pack Essentials Program (Pack Essentials).

Community Standards related to COVID-19

We are all responsible for protecting ourselves and our community. Please see the community standards (which have been updated for 2021) and Rule 04.21.01 regarding Personal Safety Requirements Related to COVID-19 RUL 04.21.01 – Personal Safety Requirements Related to COVID-19 – Policies, Regulations & Rules

Course Expectations Related to COVID-19:

- **Face Coverings:** All members of the NC State academic community are expected to follow all university policies and guidelines, including the Personal Safety Rule and community standards, for the use of face coverings. Face coverings are required in instructional spaces. Face coverings should be worn to cover the nose and mouth and be close fitting to the face with minimal gaps on the sides.
- **Course Attendance**: NC State attendance policies can be found at: REG 02.20.03 – Attendance Regulations – Policies, Regulations & Rules. Please refer to the course's attendance, absence, and deadline policies for additional details. If you are quarantined or otherwise need to miss class because you have been advised that you may have been exposed to COVID-19, you should not be penalized regarding attendance or class participation. However, you will be expected to develop a plan to keep up with your coursework during any such absences. If you become ill with COVID-19, you should follow the steps outlined in the health and participation section above. COVID 19-related

absences will be considered excused; documentation need only involve communication with your instructor.

- **Technology Requirements:** This course may require particular technologies to complete coursework. Be sure to review the syllabus for these expectations, and see the syllabus technical requirements for your course. If you need access to additional technological support, please contact the Libraries' Technology Lending Service: (Technology Lending).

Course Changes Related to COVID-19

NO LONGER AVAILABLE - Grading/Scheduling Changing Options Related to COVID-19

Two policies, enhanced S/U Grading Option and Late Drop, put in place at the beginning of the COVID-19 pandemic have been discontinued.

In some cases, an option may be to request an "incomplete" in the course. If you are experiencing difficult or extenuating circumstances, you should discuss possible options with your instructor and your academic advisor.

Need Help?

If you find yourself in a place where you need help, academically or otherwise, please review these Step-by-Step Help Topics.

Other Important Resources

- **Keep Learning:** Keep Learning
- **Protect the Pack FAQs:** Frequently Asked Questions | Protect the Pack
- **NC State Protect the Pack Resources for Students:** Resources for Students | Protect the Pack
- **Academic Success Center** (tutoring, drop in advising, career and wellness advising): Academic Success Center.
- **NC State Keep Learning, tips for students opting to take courses remotely:** Keep Learning Tips for Remote Learning
- **Introduction to Zoom for students:** https://youtu.be/5LbPzzPbYEw
- **Learning with Moodle, a student's guide to using Moodle:** https://moodle-projects.wolfware.ncsu.edu/course/view.php?id=226
- **NC State Libraries** Technology Lending Program